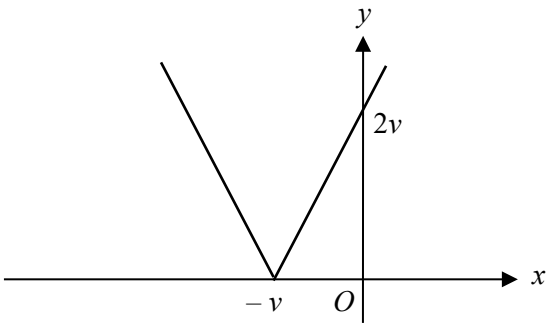
VJC 2026 H2 Math Prelim Exam Paper 2

Section A: Pure Mathematics [40 marks]

1 The diagram shows the graph of $y = |\mathrm{f}(x)|$, where $\mathrm{f}(x) = ax + b$ and a and b are constants. The graph intersects the x-axis and y-axis at $(-v, 0)$ and $(0, 2v)$ respectively, where v is a positive constant.

(a) Find the possible expressions for f(x) in terms of v. [3]



(b) It is given that $a < 0$. On separate diagrams, sketch the graphs of

(i) $y = |\mathrm{f}(2x - 2v)|$, [2]

(ii) $y = \mathrm{f}(|x|)$, [2]

labelling the points where the graphs intersect or touch the axes.

2 It is given that $P(z) = z^6 + az^4 + bz^2 + c$, where a, b and c are real constants and $c < 0$.

(a) David claims that if w is a root of $P(z) = 0$, then $-w$ must also be a root. Explain why David's claim is true. [1]

It is given $1 + \mathrm{i}$ is a root of $P(z) = 0$.

(b) State three other non-real roots of the equation $P(z) = 0$. [3]

(c) Find the remaining two roots of the equation, leaving your answers in terms of c. [2]

3(a) Referred to the origin O, points A, B and C have position vectors **a**, **b** and **c** respectively. Vectors **a**, **b** and **c** are non-zero vectors, and **a** and **c** are non-parallel. Given that M is the midpoint of AC and $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c}$, show that O, B and M are collinear. [3]

(b) It is given that **p** is a non-zero vector and **q** is a unit vector. The angle between **p** and **q** is $\cos^{-1}\left(\frac{1}{6}\right)$ and $|\mathbf{p} + 10\mathbf{q}| = 2|\mathbf{p} - 5\mathbf{q}|$. Find the magnitude of **p**. [4]

4 **(a)** A curve has equation $\ln y + xy = \cos x$.

(i) Find the Maclaurin's series for y up to and including the term in x^2. [4]

(ii) Another curve has equation $\ln y + xy = \text{cosec } x$. Explain why no Maclaurin's series for y exist. [1]

(b) By using small angle approximations, where x is small enough for x^3 and higher powers of x to be neglected, show that $\dfrac{\cos\left(x - \dfrac{3\pi}{4}\right)}{\sqrt{2-2x}} \approx \dfrac{1}{16}\left(ax^2 + bx + c\right)$, where a, b and c are integers to be determined. [4]

5 The curve C has equation

$$y = \frac{-2x^2 - 5x + a}{x - 2},$$

where a is a constant and $a < 18$.

(a) Prove, using an algebraic method, that $\dfrac{-2x^2 - 5x + a}{x - 2}$ cannot lie between two certain values, to be determined, in terms of a. [4]

It is given that $a = 0$.

(b) Find the equations of the asymptotes of C. [2]

(c) Sketch the graph of C, stating the equations of the asymptotes and the coordinates of stationary points and any point(s) where the curve crosses the axes. [3]

(d) By adding an appropriate graph to the same diagram, determine the number of real roots of the equation

$$2x^4 + 5x^3 + 2x - 4 = 0.$$ [2]

Section B: Probability and Statistics [60 marks]

6 A factory produces a large batch of electronic microchips, of which 20% are defective. A quality control test consists of an inspector selecting a random sample of n microchips from the batch and counting the number that are defective. If at most one defective microchip is found, the batch is accepted. If more than two defective microchips are found, the batch is rejected. If exactly two defective microchips are found, the inspector selects a second random sample of n microchips and the batch is rejected if this sample contains any defective microchips. Otherwise, the batch is accepted.

(a) Show that the probability that a randomly chosen batch is accepted is P, where

$$P = (0.8 + 0.2n)(0.8)^{n-1} + 0.02n(n-1)(0.8)^{2n-2} .$$ [3]

(b) Hence, find the least possible value of n, given that the probability that a randomly chosen batch is accepted is less than 0.1. [2]

(c) Given instead that $P = 0.398$, find the probability that out of 100 randomly selected batches that are inspected, between 40 and 45 batches inclusive are accepted. [2]

7 For events A and B, it is given that $\mathrm{P}(A \cap B) = \frac{7}{20}$, $\mathrm{P}(B|A) = \frac{7}{12}$ and $\mathrm{P}(A|B) = \frac{7}{9}$.

(a) Show that $\mathrm{P}(A \cap B') = \frac{1}{4}$. [2]

(b) Find $\mathrm{P}(A \cup B)$. [2]

(c) Find $\mathrm{P}(A' \cap B')$. [1]

It is given that $\mathrm{P}(C) = \frac{7}{10}$, where C is an event such that events A and C are independent, and B and C are independent.

(d) Find the greatest and least possible values of $\mathrm{P}(A \cap B \cap C)$. [3]

8 A box contains 6 red balls, 5 white balls and 3 black balls. In a game, a player draws one ball at random from the box at a time, without replacement. The game ends as soon as the player has drawn two balls of the same colour. The random variable X denotes the total number of balls the player draws at the end of the game.

(a) Show that $\mathrm{P}(X=3)=\frac{81}{182}$ and tabulate the probability distribution of X. [4]

The player receives prize money of $\$(200-50X)$ at the end of the game.

(b) Find the expectation and variance of the prize money received by the player at the end of the game, leaving your answer correct to 2 decimal places. [3]

If the first game ends after exactly two balls are drawn, the player plays a second game. Before the second game begins, all the balls are returned to the box, and the second game is played according to the same rules as the first. A player plays at most two games.

(c) Find the expected total prize money received by a player over all games. [2]

While waiting for the game to start, five players are to be seated in a row of 10 seats.

(d) Find the number of distinct seating arrangements if two particular players are seated next to each other. [2]

9 **In this question you should state clearly the values of the parameters of any distribution you use.**

An engineering team conducts analyses on the capacities of two types of battery cells, Standard and Premium. The capacity of each battery cell is measured in mAh. The capacity of a Standard battery cell has distribution $\mathrm{N}(\mu, \sigma^2)$. A Standard battery cell is equally likely to have a capacity of less than 2750 mAh and more than 3650 mAh.

(a) State the value of μ. [1]

It is given that the probability that the capacity of a Standard battery cell is between 2750 mAh and 3650 mAh is 0.860.

(b) Show that $\sigma = 305,$ correct to 3 significant figures. [2]

(c) Sketch the distribution of the capacity of a Standard battery cell for capacities between 1900 mAh and 4500 mAh. [2]

A Premium battery is created by optimising a Standard battery cell so that its capacity is increased by 10%.

(d) Find the probability that the total capacities of two Standard battery cells exceed the capacity of a Premium battery cell by more than 3000 mAh. [3]

The company has recently introduced a third type of battery cell, Budget. The capacity of a randomly selected Budget battery cell has mean 1200 mAh and standard deviation 400 mAh. Three independent random samples of 100 Budget battery cells each are selected.

(e) Estimate the probability that exactly two of the three samples have mean capacity of less than 1150 mAh. [3]

[Turn over

10 Bottles of tomato juice produced by a company are said to contain on average 250 ml of tomato juice. After receiving feedback from some consumers regarding the volume of tomato juice per bottle, the production manager took a random sample of n bottles and measured the volume of tomato juice in each bottle, x ml, for quality control purposes.

(a) State what it means for a sample to be random in this context. [1]

(b) In a random sample where $n = 12$, the values of x are recorded as follows.

252, 245, 251.8, 249.2, 245, 250.5, 245.5, 239.5, 248.5, 250.2, 245, 249.6

(i) Taking the population standard deviation to be 5 ml, test, at the 5% significance level, whether the mean volume has been overstated. State a necessary assumption for the validity of your test. [5]

(ii) Explain the meaning of the p-value found in part **(b)(i)** in the context of this question. [1]

The company installs a new machine to produce bottles of prune juice with mean volume k ml. A random sample of 60 bottles of prune juice is taken, and the mean and variance of these bottles are found to be 179.3 ml and 50.8 ml^2 respectively.

(c) Calculate the set of values of k for the test to conclude, at 2% level of significance, that the mean volume of prune juice in a bottle is not k ml. [4]

11 A marketing team is interested in finding out how the number of new user acquisitions (y, in hundreds) depends on the amount spent on digital advertising (x, in hundreds of dollars). A study is done which produced the following data shown in the table.

x	300	1100	1900	2700	3500	4300	5100	5900	6700	7500
y	55	66	70	73	76	k	79	80	81	82

(a) Given that the regression line of y on x is $y = 0.0031667x + 61.65$, show that $k = 78$. [3]

(b) Sketch a scatter diagram of the data. [1]

(c) Explain from your scatter diagram why the relationship between x and y should not be modelled by an equation of the form $y = ax + b$, where a and b are constants. [1]

(d) Suppose that the relationship between x and y is modelled by the equation of the form $\ln y = c + d \ln x$, where c and d are constants. Find the values of c and d and the value of the product moment correlation coefficient for this model [3]

(e) Use the model $\ln y = c + d \ln x$, with the values of c and d found in part **(d)**, to estimate the amount spent on digital advertising when the number of user acquisitions is 6000. (You should give your answer correct to the nearest hundred.) Give two reasons why your estimate is reliable. [3]

(f) Dilip reviews the data and concludes: "The strong positive correlation proves that a higher amount spent on digital advertising is the direct cause of the increase in new user acquisition." State, with a reason, whether Dilip's conclusion is valid. [1]